

INFLATION

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THE ROAD TOWARDS 2% INFLATION

12-month PCE inflation in July 2026:

- ▶ Headline: 3.7%
- ▶ Ex Energy Goods: 3.3%
- ▶ Core: 3.3%

Annualized PCE inflation through the year in July 2026:

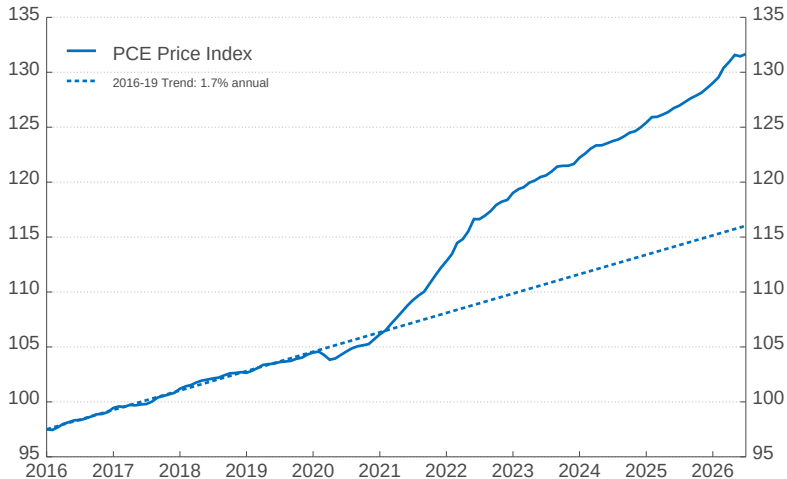
- ▶ Headline: 4.1%
- ▶ Ex Energy Goods: 3.6%
- ▶ Core: 3.7%

PCE Ex Energy Goods inflation trend for 2023-26: 2.9%

Prices and Inflation

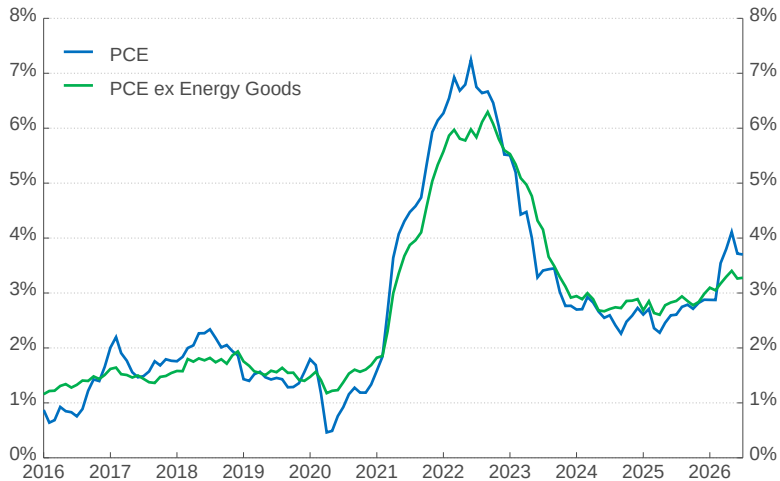
PRICE LEVEL

PRICES DECLINE EARLY IN THE PANDEMIC AND RISE PERSISTENTLY ABOVE THEIR TREND AFTERWARDS



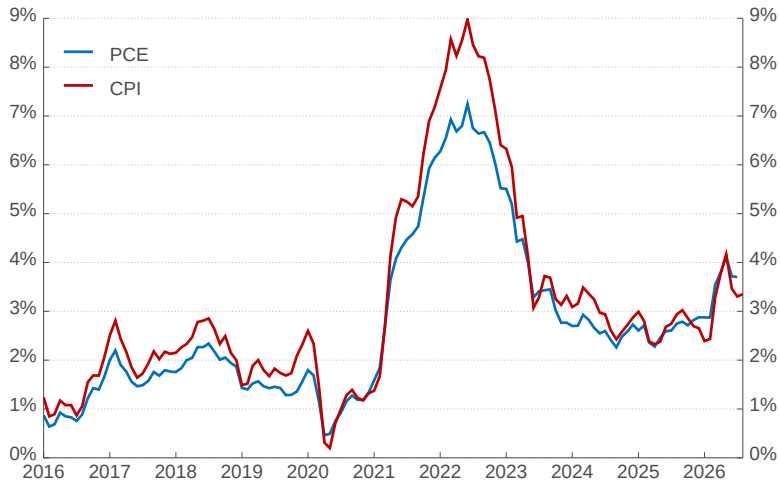
INFLATION

INFLATION HAS REMAINED ABOVE 2% SINCE MARCH 2021—DYNAMICS AMPLIFIED BY ENERGY ▷ HISTORICAL SERIES



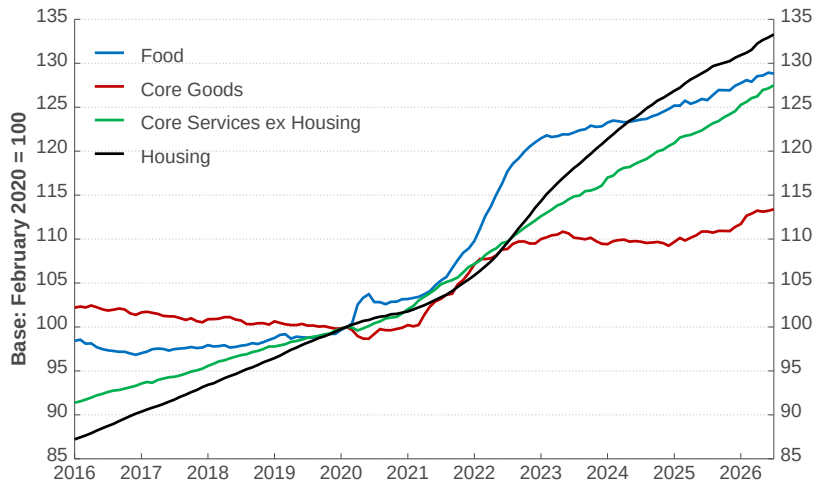
COMPARING DYNAMICS IN PCE AND CPI

▷ EXCLUDING ENERGY



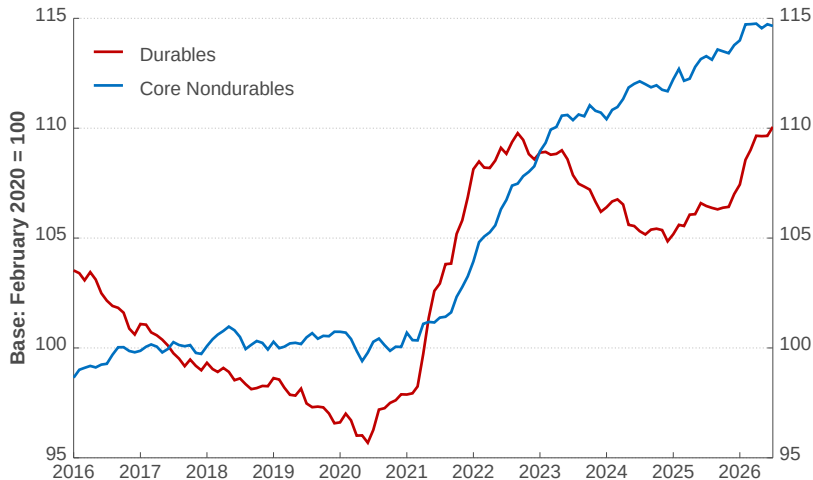
COMPONENTS OF PCE PRICE INDEX

FOOD, CORE GOODS, CORE SERVICES EX HOUSING, AND HOUSING



CORE GOODS: DURABLES VS. NONDURABLES

EFFECT OF TARIFFS EVIDENT ON PRICES OF CORE DURABLE GOODS—CORE NONDURABLES GROWING STEADILY



INFLATION: AGGREGATE MEASURES

▷ CPI

Annualized PCE Inflation Rates

	All Goods & Services	Ex Energy Goods	Ex Food & Energy
2016–19	1.7%	1.6%	1.7%
2020	1.3%	1.7%	1.5%
2021	6.1%	5.3%	5.2%
2022	5.5%	5.6%	5.0%
2023	2.8%	2.9%	3.1%
2024	2.7%	2.9%	3.0%
2025	2.9%	3.0%	3.0%
2026 YTD	4.1%	3.6%	3.7%

SOURCES: Bureau of Economic Analysis and author's calculations.

NOTE: The 2026 period is the year through July.

INFLATION: MAIN COMPONENTS

▷ CPI

Annualized PCE Inflation Rates by Categories

	Food	Energy	Core Goods	Core Services ex Housing	Housing
2016–19	0.2%	4.2%	−0.6%	2.2%	3.4%
2020	4.0%	−7.5%	0.1%	2.0%	2.2%
2021	5.6%	30.3%	6.2%	5.2%	3.7%
2022	11.1%	6.7%	3.2%	4.9%	7.8%
2023	1.5%	−2.1%	0.0%	3.5%	6.4%
2024	1.6%	−1.0%	−0.2%	3.9%	4.7%
2025	2.1%	2.2%	2.0%	3.3%	3.3%
2026 YTD	1.9%	18.9%	3.1%	4.1%	3.5%

SOURCES: Bureau of Economic Analysis and author's calculations.

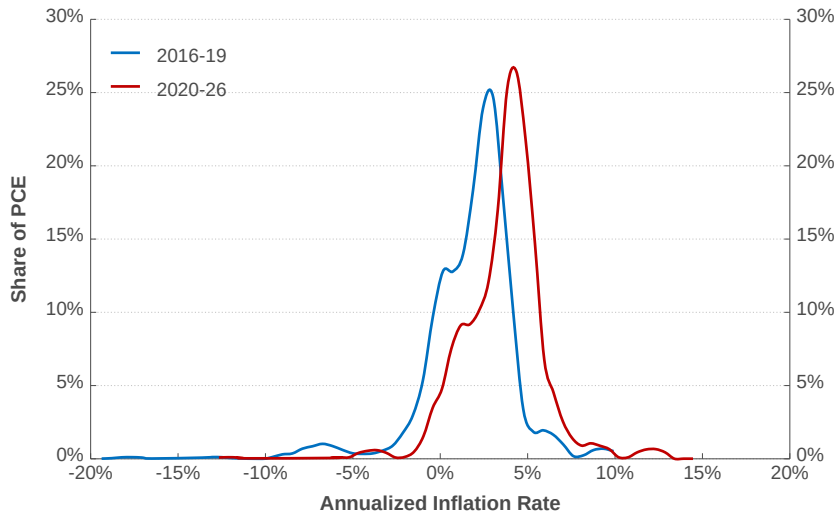
NOTE: The 2026 period is the year through July.

Inflation Distributions

INFLATION INCREASES HAVE BEEN WIDESPREAD

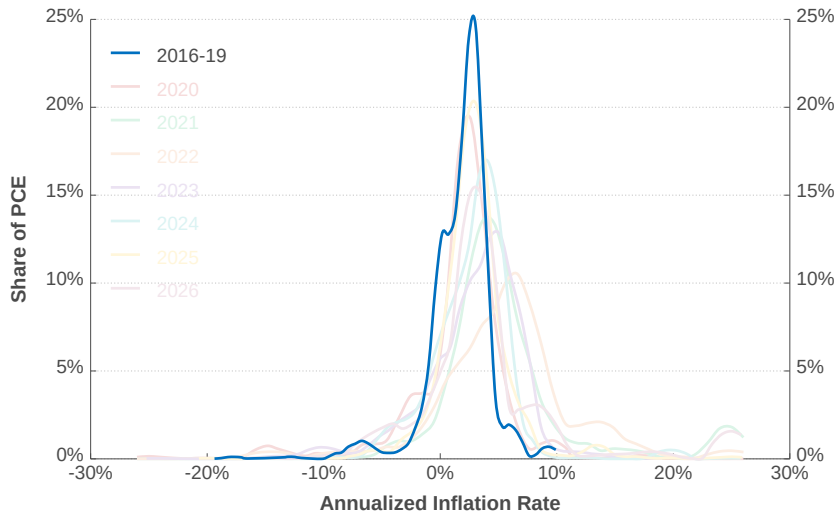
► METHODOLOGY

ESTIMATED DISTRIBUTION OF PCE INFLATION ► CORE



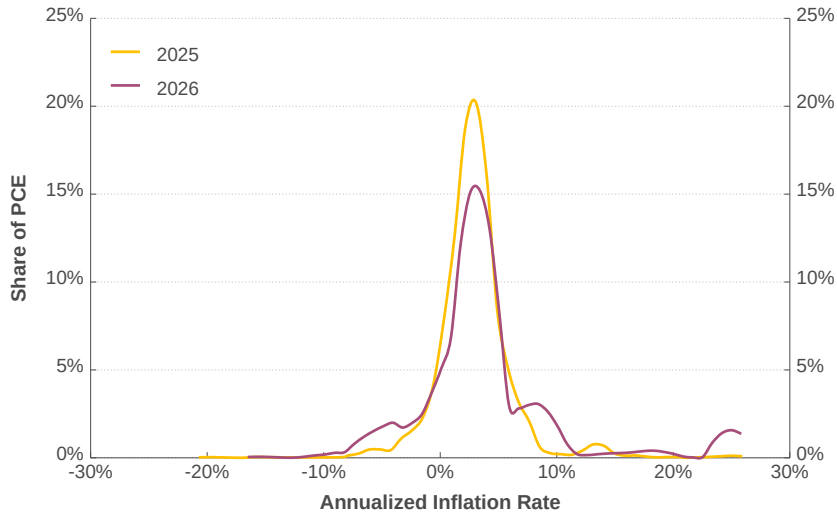
INFLATION DISTRIBUTIONS: PROGRESSION FROM 2016-19 TO 2026

ESTIMATED DISTRIBUTION OF PCE INFLATION



INFLATION DISTRIBUTIONS: 2025 vs. 2026

ESTIMATED DISTRIBUTION OF PCE INFLATION ▸ CORE

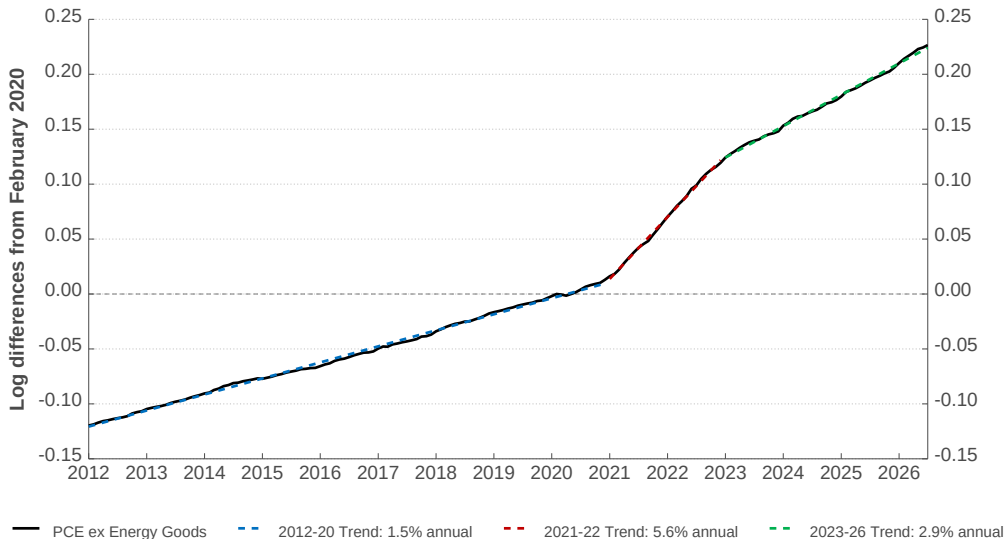


Inflation Regimes

PRICE LEVEL DYNAMICS: THREE REGIMES

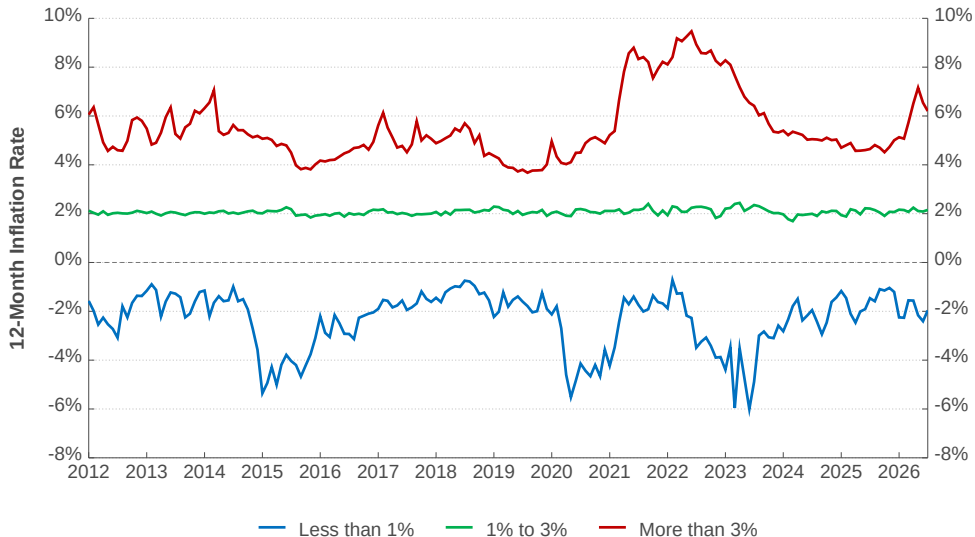
[► BLOG](#)

(1) BELOW-TARGET INFLATION: 2012-20 (2) GREAT INFLATION: 2021-22 (3) ABOVE-TARGET INFLATION: 2023-26



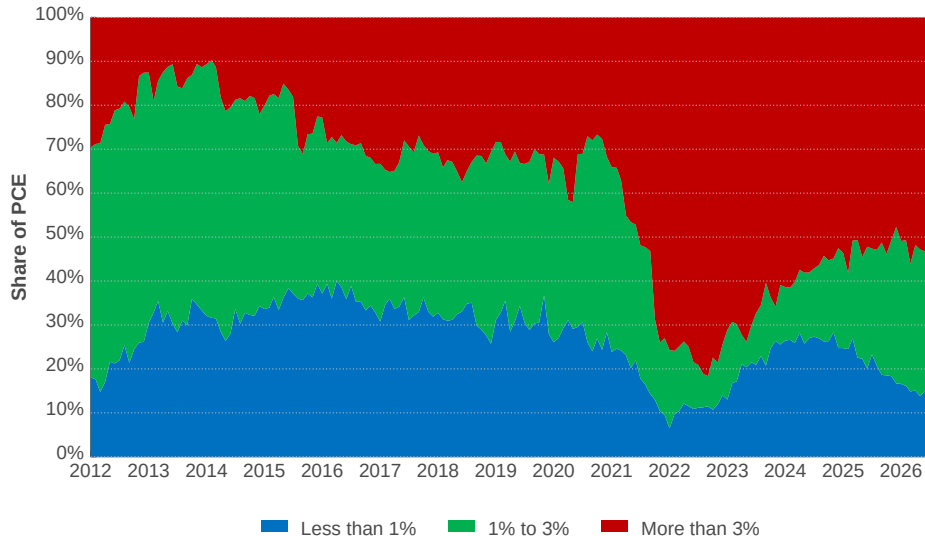
DYNAMIC PCE 12-MONTH INFLATION

PRODUCTS REGROUPED MONTHLY BY THEIR CURRENT 12-MONTH INFLATION ▸ EX ENERGY GOODS



DYNAMIC PCE EXPENDITURE SHARES

PRODUCTS REGROUPED MONTHLY BY THEIR CURRENT 12-MONTH INFLATION ▸ EX ENERGY GOODS



AVERAGE PCE EXPENDITURE SHARES

PRODUCTS REGROUPED MONTHLY BY THEIR CURRENT 12-MONTH INFLATION ▶ EX ENERGY GOODS

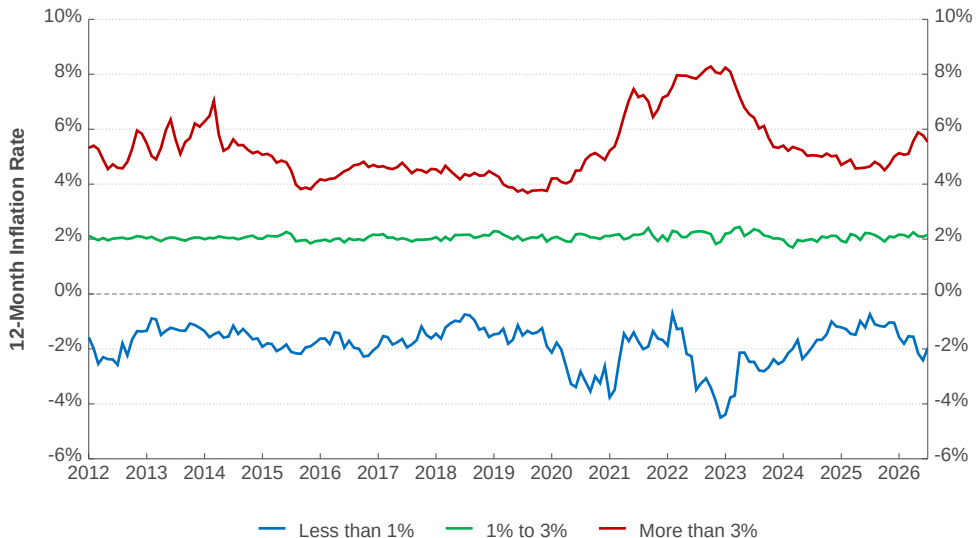
Average Shares of Nominal PCE

	Less than 1%	1% to 3%	More than 3%
2016–19	33.2%	35.6%	31.3%
2020	27.7%	40.1%	32.2%
2021	18.3%	30.3%	51.5%
2022	11.0%	11.9%	77.2%
2023	20.9%	11.6%	67.5%
2024	26.7%	16.1%	57.3%
2025	21.4%	26.1%	52.4%
2026	15.5%	31.4%	53.2%

NOTE: Arithmetic averages of monthly expenditure shares. 2026 covers January–July.

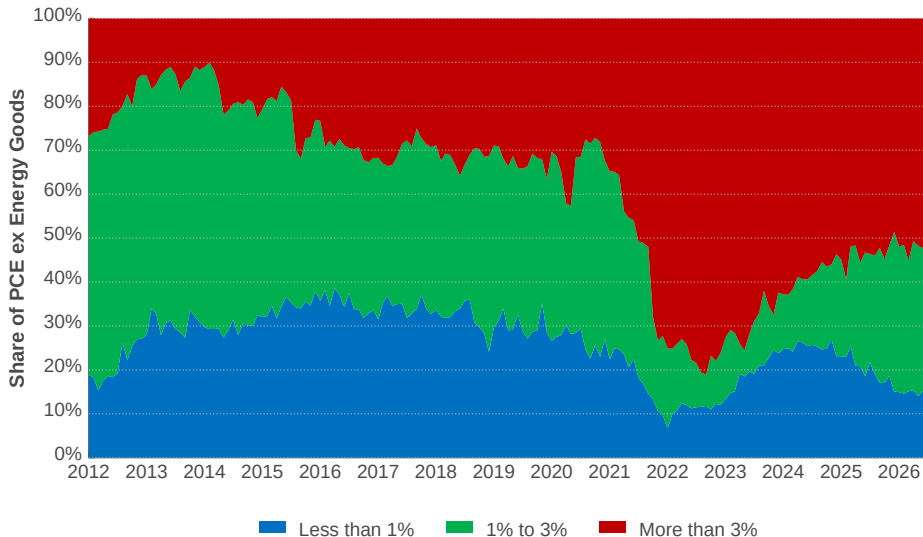
DYNAMIC PCE ex ENERGY GOODS 12-MONTH INFLATION

PRODUCTS REGROUPED MONTHLY BY THEIR CURRENT 12-MONTH INFLATION ▸ TOTAL PCE



DYNAMIC PCE ex ENERGY GOODS EXPENDITURE SHARES

PRODUCTS REGROUPED MONTHLY BY THEIR CURRENT 12-MONTH INFLATION ▸ TOTAL PCE



AVERAGE PCE ex ENERGY GOODS EXPENDITURE SHARES

PRODUCTS REGROUPED MONTHLY BY THEIR CURRENT 12-MONTH INFLATION ▸ TOTAL PCE

Average Shares of Nominal PCE ex Energy Goods

	Less than 1%	1% to 3%	More than 3%
2016–19	32.9%	36.4%	30.8%
2020	26.8%	40.9%	32.4%
2021	18.5%	30.8%	50.7%
2022	11.1%	12.2%	76.7%
2023	19.4%	11.4%	69.2%
2024	25.2%	16.2%	58.6%
2025	20.0%	26.5%	53.5%
2026	15.2%	32.1%	52.7%

NOTE: Arithmetic averages of monthly expenditure shares. 2026 covers January–July.

LONG-TERM INFLATION EXPECTATIONS

MARKETS VS. MODELS VS. FORECASTERS

10-Year Inflation Expectations — Period Averages

	TIPS	Swaps	Cleveland	DKW	SPF
2012–20	1.9%	2.2%	1.7%	2.2%	2.0%
2021–22	2.4%	2.7%	1.9%	2.2%	2.3%
2023–26	2.3%	2.5%	2.3%	2.7%	2.2%
Change 2012-20 to 2023-26	0.4%	0.3%	0.6%	0.4%	0.2%

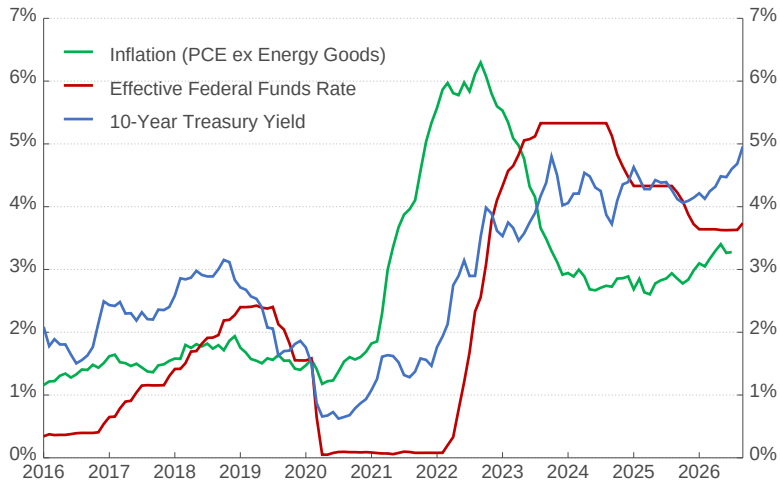
SOURCES: FRED, Federal Reserve Board, Federal Reserve Bank of Philadelphia, Bloomberg and author's calculations.

NOTE: The 2026 period is the year through July.

Interest Rates

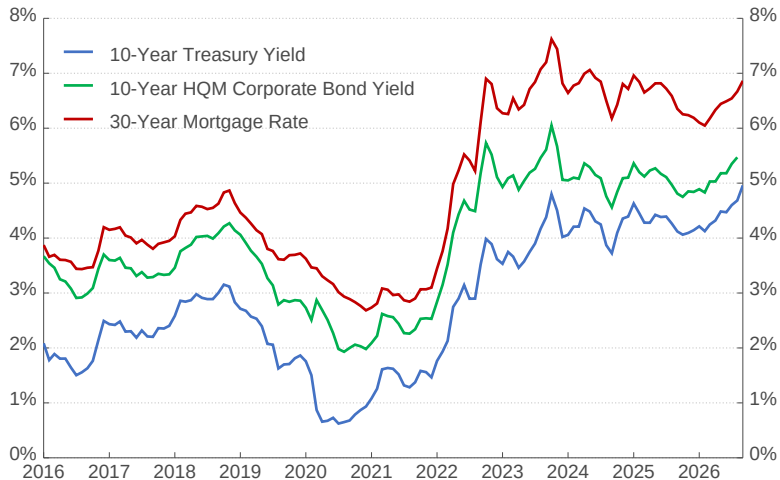
MONETARY POLICY

EASING CYCLES IN SEPTEMBER-DECEMBER 2024 AND SEPTEMBER-DECEMBER 2025



MONETARY POLICY AND LONG-TERM RATES

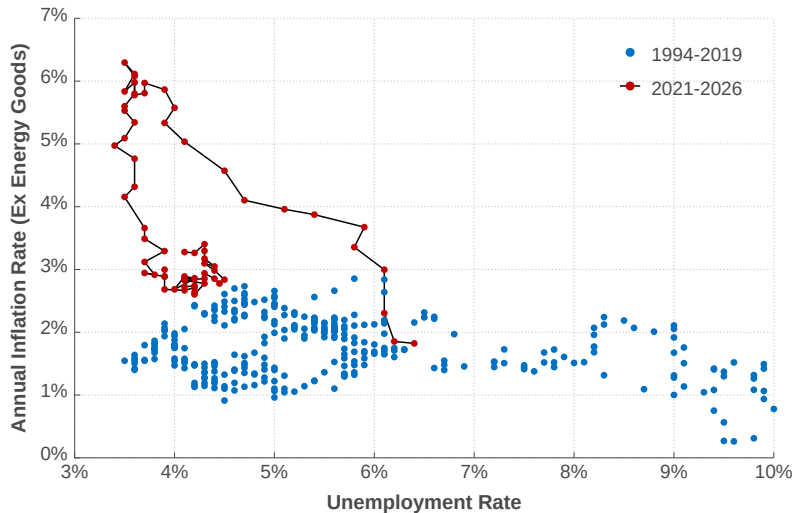
ANNUAL RATES



Phillips Curve

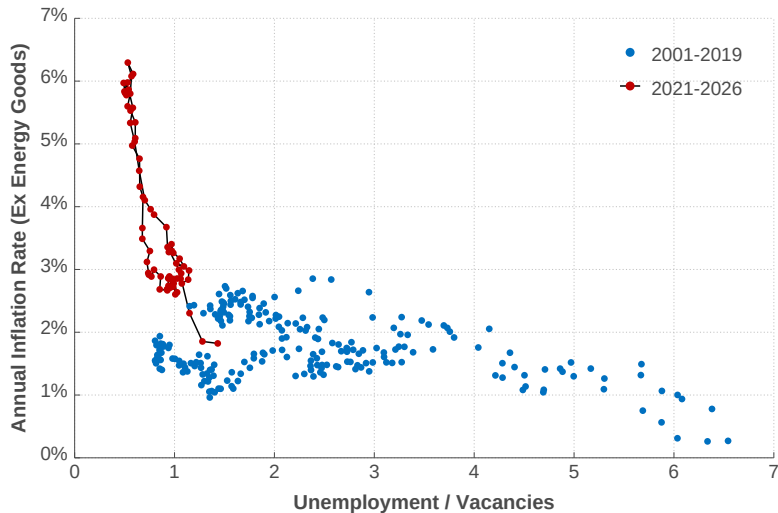
THE PHILLIPS CURVE

INFLATION VS. UNEMPLOYMENT



ALTERNATIVE PHILLIPS CURVE

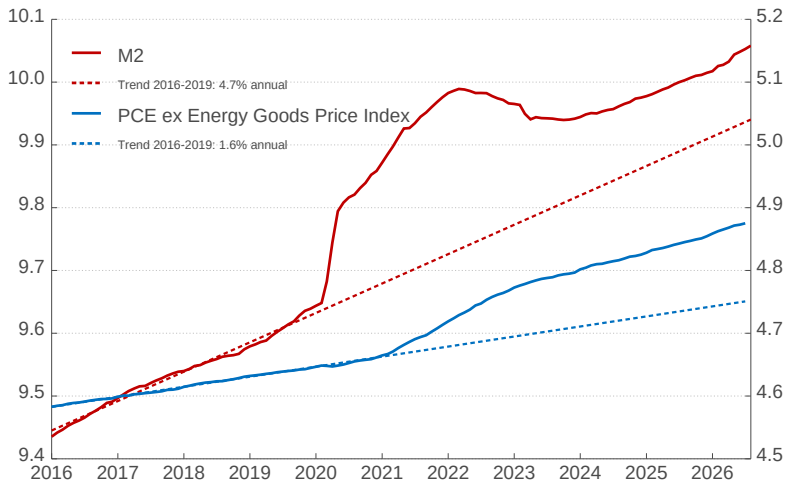
INFLATION VS. UNEMPLOYMENT-TO-VACANCIES RATIO



$$\mathbf{M} \times \mathbf{V} = \mathbf{P} \times \mathbf{Y}$$

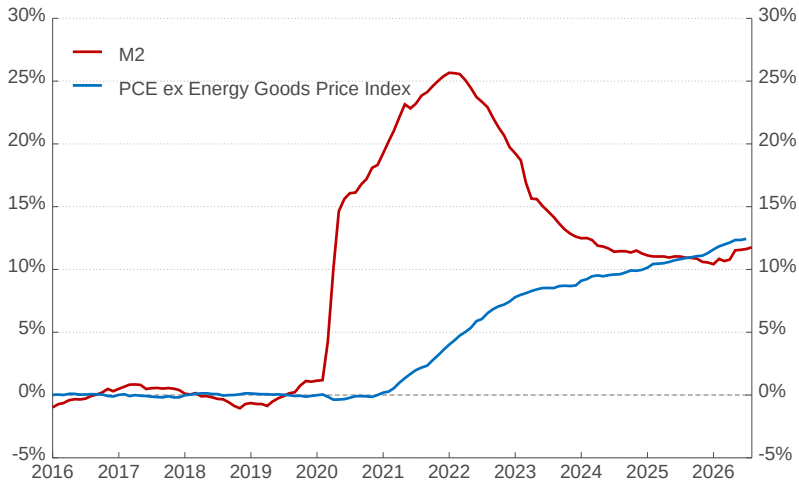
MONEY VS PRICES: MONETARY EXPANSION DURING COVID

IN LOGS: M2 ON LEFT AXIS, PRICES ON RIGHT AXIS—M2 GREW AT SAME RATE AS DISPOSABLE INCOME PRE-COVID



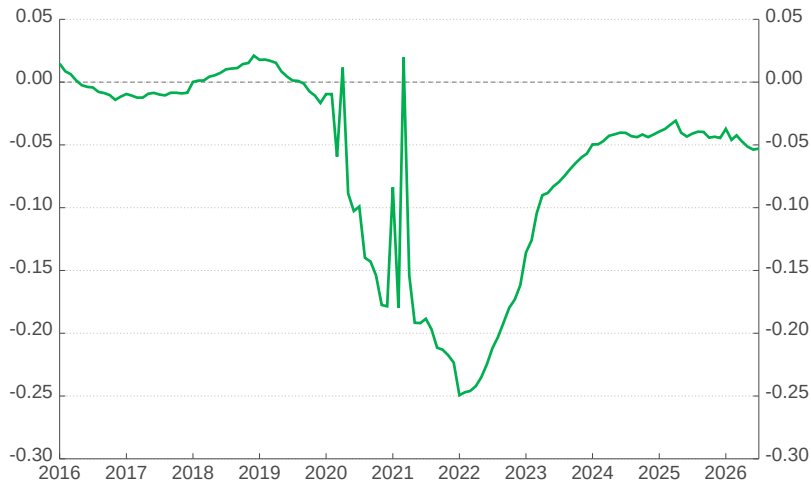
PRICE INCREASE VS. MONETARY EXPANSION

DEVIATIONS FROM PRE-COVID TREND



VELOCITY

$\text{LN VELOCITY} = \text{LN M2} - \text{LN DISPOSABLE PERSONAL INCOME}$ (BASE: 2016-19 = 0)



Appendix

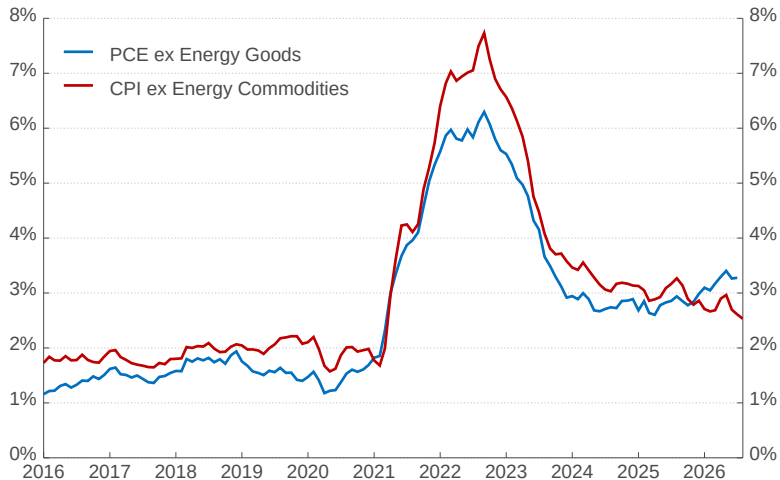
CURRENT INFLATION IN HISTORICAL CONTEXT

PCE EX ENERGY GOODS ANNUAL INFLATION [▷ BACK](#)



PCE ex ENERGY GOODS AND CPI ex ENERGY COMMODITIES

▷ [BACK](#)



CPI INFLATION AGGREGATES

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Annualized CPI Inflation Rates

	All Goods & Services	Ex Energy Commodities	Ex Food & Energy
2016–19	2.1%	1.9%	2.1%
2020	1.3%	2.0%	1.6%
2021	7.2%	5.7%	5.5%
2022	6.4%	6.7%	5.7%
2023	3.3%	3.6%	3.9%
2024	2.9%	3.1%	3.2%
2025	2.7%	2.9%	2.6%
2026 YTD	3.7%	3.4%	2.7%

SOURCES: U.S. Bureau of Labor Statistics and author's calculations.

NOTE: The 2026 period is the year through August.

CPI INFLATION COMPONENTS

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Annualized CPI Inflation Rates by Categories

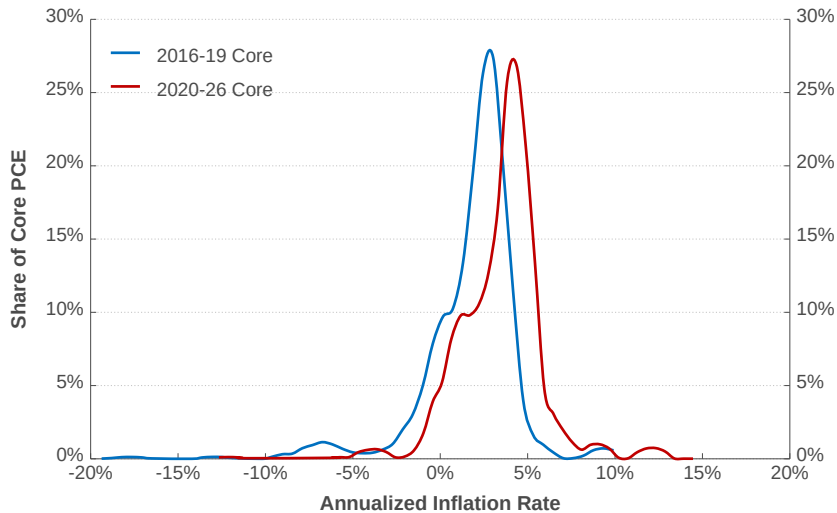
	Food	Energy	Core Commodities	Services Less Rent	Rent of Shelter
2016–19	1.2%	3.9%	−0.2%	2.2%	3.3%
2020	3.9%	−7.0%	1.6%	1.4%	1.9%
2021	6.0%	30.3%	10.7%	3.9%	4.2%
2022	10.1%	6.9%	2.1%	7.5%	7.6%
2023	2.7%	−1.9%	0.1%	3.5%	6.2%
2024	2.4%	−0.3%	−0.7%	4.0%	4.6%
2025	3.0%	2.1%	1.4%	3.4%	3.1%
2026 YTD	2.4%	19.2%	0.6%	3.4%	3.3%

SOURCES: U.S. Bureau of Labor Statistics and author's calculations.

NOTE: The 2026 period is the year through August.

CORE INFLATION INCREASES HAVE BEEN WIDESPREAD

ESTIMATED DISTRIBUTION OF CORE PCE INFLATION [▷ BACK](#)



CORE INFLATION DISTRIBUTIONS: 2025 vs. 2026

ESTIMATED DISTRIBUTION OF CORE PCE INFLATION [▷ BACK](#)

